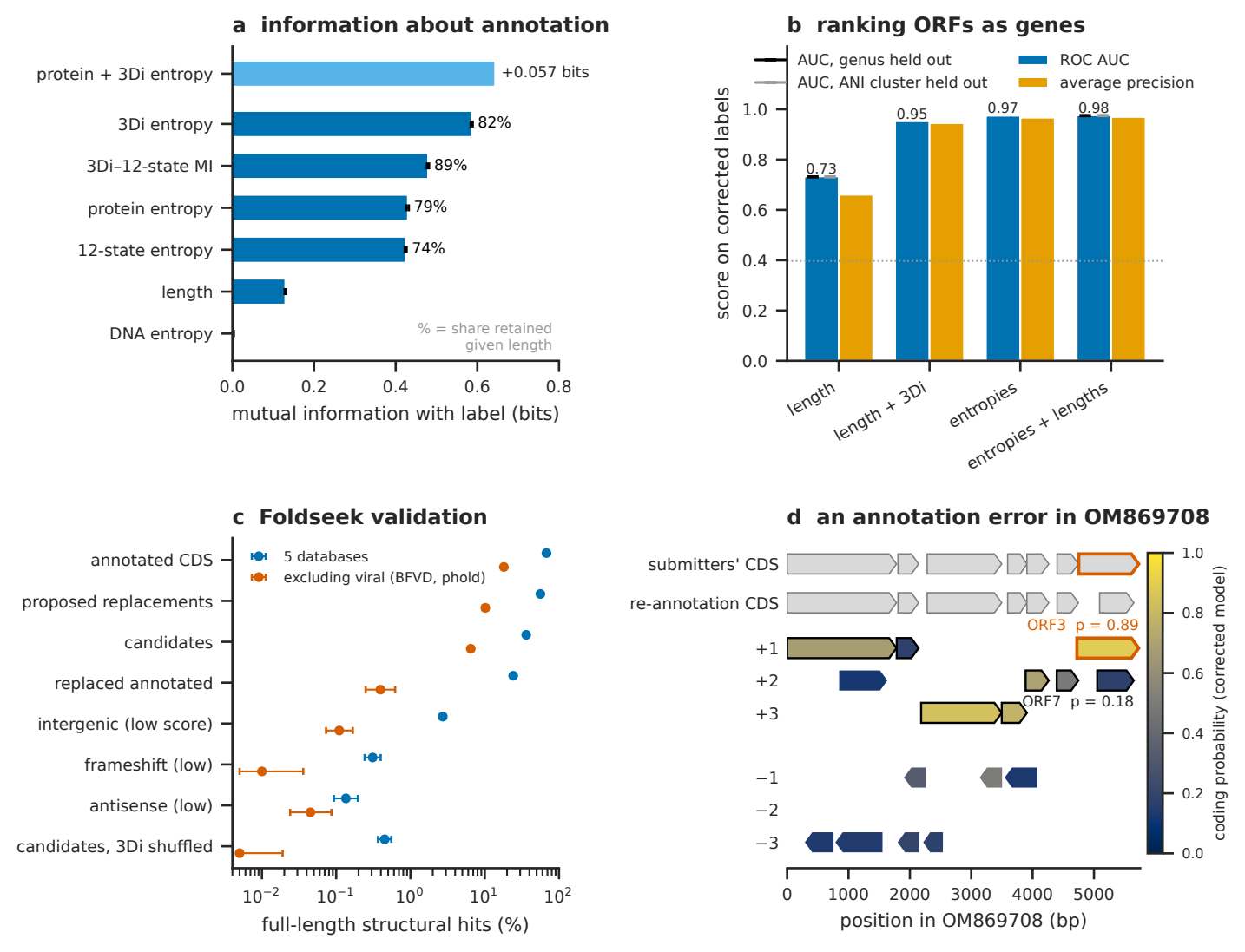
a information about annotation
protein + 3Di entropy
+0.057 bits
3Di entropy
82%
3Di–12-state MI
89%
protein entropy
79%
12-state entropy
74%
length
DNA entropy
% = share retained given length
0.0
0.2
0.4
0.6
0.8
mutual information with label (bits)
b ranking ORFs as genes
AUC, genus held out
AUC, ANI cluster held out
ROC AUC
average precision
score on corrected labels
0.0
0.2
0.4
0.6
0.8
1.0
0.73
0.95
0.97
0.98
length
length + 3Di
entropies
entropies + lengths
c Foldseek validation
5 databases
excluding viral (BFVD, phold)
annotated CDS
proposed replacements
candidates
replaced annotated
intergenic (low score)
frameshift (low)
antisense (low)
candidates, 3Di shuffled
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
full-length structural hits (%)
d an annotation error in OM869708
submitters' CDS
re-annotation CDS
ORF3 p = 0.89
+1
+2
+3
ORF7 p = 0.18
−1
−2
−3
0
1000
2000
3000
4000
5000
position in OM869708 (bp)
coding probability (corrected model)
0.0
0.2
0.4
0.6
0.8
1.0